

Artificial Intelligence - MCQ - Slot C

Total points 20/20 ?

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MCQ

20 of 20 points

C02-AT1

✓ A thermostat monitors room temperature (input) and switches the heater *1/1 on or off (output) accordingly, without maintaining any internal record of past states.

Which type of agent does this best represent?

- ☐ Goal-Based Agent
- ☐ Utility-Based Agent
- ☒ Simple Reflex Agent
- ☐ Learning Agent



✓ In A* search, the evaluation function is calculated using both the path cost and heuristic value. Suppose a node has a path cost of 7 and a heuristic estimate of 5. What is the value of $f(n)$? *1/1

☒ 12



☐ 2

☐ 35

☐ 7

✓ A rescue robot searches every room of a collapsed building. Every movement costs the same. *1/1

If the nearest victim must be found first, which search strategy is appropriate?

☐ DFS

☒ BFS



☐ Hill Climbing

☐ Stimulated Annealing



✓ Why is 'IDS' (Iterative Deepening Search) often preferred for large search spaces where the goal depth is unknown, even if memory is not the main constraint? *1/1

- ☐ It handles infinite paths better.
- ☐ It reduces the branching factor.
- ☐ It is faster than A*.
- ☒ It combines BFS optimally and DFS space



✓ A search agent is trying to find the shortest path in a graph. It uses A* with $h(n) = 0$. This configuration is equivalent to: *1/1

- ☐ Breadth-First Search.
- ☒ Uniform-Cost Search.
- ☐ Greedy Best-First Search.
- ☐ Depth-First Search.



✓ A company uses Hill Climbing to optimize delivery schedules. The algorithm repeatedly stops before finding the best schedule. *1/1

What is the MOST likely reason?

- ☒ Local optimum
- ☐ Infinite heuristic
- ☐ Complete search tree
- ☐ Breadth expansion



✓ Why is 'Stochastic Hill Climbing' often superior to standard 'Steepest-Ascent Hill Climbing' in complex landscapes? *1/1

- ☒ It chooses randomly among uphill moves, where the probability of selection can vary with the steepness of the move, reducing susceptibility to local maxima traps. ✓
- ☐ It guarantees finding global optima in polynomial time.
- ☐ It eliminates the need for neighbor generation functions.
- ☐ It evaluates global state graphs using priority queue limits.

✓ You are building a search heuristic for an 8-puzzle. You have two heuristics: h_1 (number of misplaced tiles) and h_2 (total Manhattan distance). *1/1

Why is h_2 generally preferred over h_1 in A* search?

- ☐ It is easier to compute.
- ☐ It is more consistent.
- ☐ It handles illegal moves.
- ☒ It dominates h_1 ✓



✓ A researcher claims that a machine can be considered intelligent if a human evaluator cannot reliably distinguish its responses from those of another human during a text-based conversation. *1/1

Which foundational concept does this describe?

- ☐ Rational Agent Model
- ☒ Turing Test
- ☐ Chinese Room Argument
- ☐ Expert System Approach



✓ In a minimax tree, each node has 3 children and the depth of the tree is 3. *1/1

The total number of leaf nodes depends on the branching factor raised to the depth.

How many leaf nodes are present?

- ☐ 9
- ☐ 12
- ☐ 16
- ☒ 27



✓ A self-driving car detects a pedestrian unexpectedly crossing the road. * 1/1

What should a rational agent do?

- ☒ Stop or safely avoid the pedestrian while minimizing overall risk.
- ☐ Continue at the same speed.
- ☐ Ignore the pedestrian.
- ☐ Wait for another vehicle.



- ✓ A warehouse robot always moves toward the shelf that appears closest to the destination using straight-line distance but sometimes gets trapped behind obstacles. *1/1

Why does this happen?

- ☒ Greedy Search ignores actual path cost. ✓
- ☐ BFS is incomplete.
- ☐ DFS uses heuristics.
- ☐ Uniform Cost Search uses heuristics.

- ✓ An autonomous delivery robot operates inside a hospital. It detects patients, avoids obstacles, changes its path when corridors are blocked, and delivers medicines on time. *1/1

Which characteristic best demonstrates that the robot behaves as a **rational agent**?

- ☐ It always follows the shortest path.
- ☒ It chooses actions that maximize successful delivery based on current information. ✓
- ☐ It memorizes every hospital corridor.
- ☐ It never changes its route.



✓ A heuristic function sometimes overestimates the actual cost to reach the goal. This violates the admissibility condition required for optimal solutions in A* search. *1/1

What happens in such a case?

- ☐ A* is guaranteed to find the optimal solution.
- ☒ A* may find a solution faster, but it is not guaranteed to be optimal. ✓
- ☐ A* behaves exactly like Uniform Cost Search.
- ☐ A* will fail to find any solution.

✓ A self-driving car has access only to its sensor data and cannot know about a sudden pedestrian hidden behind a parked truck, yet it still makes the best possible decision given what it knows. *1/1

This illustrates that rationality depends on:

- ☐ Omniscience
- ☐ The performance measure only
- ☒ Percept sequence and available knowledge, not omniscience ✓
- ☐ Random action selection



- ✓ A drone is assigned to inspect wind turbines. During flight, strong winds change its position, birds suddenly appear, and GPS signals occasionally disappear. *1/1

Which environment classification is MOST appropriate?

- ☐ Fully observable, deterministic, static
- ☒ Partially observable, stochastic, dynamic ✓
- ☐ Deterministic, static, discrete
- ☐ Fully observable, sequential

- ✓ An ambulance navigation system considers both: * 1/1

Distance already travelled Estimated remaining distance

Which search algorithm naturally supports this?

- ☐ BFS
- ☐ DFS
- ☐ Greedy Search
- ☒ A* ✓



✓ A self-driving car must plan a route through a city. The environment is 'dynamic' because other cars move independently. What is the most significant challenge this poses for a 'Problem-Solving Agent' using classical search? *1/1

- ☐ Heuristics become inadmissible.
- ☐ Branching factor is infinite.
- ☒ The solution is not executable
- ☐ The state space is unknown



✓ An agricultural robot waters crops. *1/1

It measures soil moisture, weather forecasts, and water availability before deciding.

Which agent architecture best fits?

- ☐ Simple Reflex Agent
- ☐ Goal-Based Agent
- ☒ Utility-Based Agent
- ☐ Table-Driven Agent



✓ An airline wants to assign thousands of flights to airport gates while minimizing passenger walking distance. *1/1

Which AI technique is most suitable?

- ☐ BFS
- ☐ DFS
- ☒ Local Search
- ☐ Bidirectional Search



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